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Optimizing thermal stress distribution in heat-pipe-cooled microreactors using multi-physics data-driven methods

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ABSTRACT

Heat-pipe-cooled microreactors represent cutting-edge nuclear energy systems, but they are faced with significant design challenges. The neutronic–thermal–mechanical interactions within these reactors are complex, and the excessive thermal stress in the core monolith threatens the structural integrity. We present an advanced multi-physics coupling method combining high-fidelity neutronic, thermal-hydraulic, and mechanical simulations. The model employs Monte Carlo neutron transport, finite element solid thermodynamics, and a heat pipe wall temperature model, coupled using Picard iteration. Leveraging this multi-physics coupling method, we have developed data-driven models utilizing artificial neural networks, which reduced each calculation time from 6 h to 4 min without sacrificing accuracy. To reduce the peak stress in the monolith, the non-dominated sorting genetic algorithm (NSGA-II) is applied to optimize key design parameters, such as fuel enrichment, active core height, and fuel rod pitch. An innovative heat-pipe-type optimization strategy was formulated, facilitating effective redistribution of core temperature and substantially reducing thermal stress without undermining key safety parameters or reactor performance. The optimization results showcased a 59% reduction in peak stress, with the maximum value decreasing to 68 MPa, thereby ensuring the stress levels remain within safe limits. This work suggests a viable solution for managing the thermal stress issues in heat-pipe-cooled microreactors.

1. Introduction

Microreactors are becoming increasingly prominent in the modern energy sector, playing a crucial role in terrestrial, maritime, and space exploration applications [1–5]. These compact reactors offer significant advantages, including simplified structures, high reliability, and adaptability, making them ideal for distributed energy systems, remote power supplies, and emergency power scenarios.

The heat-pipe-cooled microreactor (heat pipe reactor) has stood out as a promising concept of microreactor, initially developed at Los Alamos National Lab (LANL) [6]. In a heat pipe reactor, heat pipes are integrated into the monolithic core in a lattice configuration, offering a novel and distinctive approach to remove fission and decay heat. Each heat pipe contains a minimal quantity of working fluid within a hermetically sealed steel tube [7]. The system operates passively, maintaining internal pressures below atmospheric levels. The primary function of these heat pipes is to transfer thermal energy from the evaporator section located within the core to the condenser situated outside the core efficiently. This is achieved through a continuous flow of vapor and liquid, which maintains a constant temperature throughout the process [8]. The entire process relies on the physical properties of the system to achieve efficient, non-powered heat transfer.

The employment of passive heat transfer through heat pipes in place of the primary loop system in conventional reactors eliminates the need for primary piping, mechanical circulation pumps, and auxiliary support systems. This design choice gives rise to a number of advantageous characteristics of heat pipe reactors, including structural simplicity, improved reliability, inherent passive heat transfer mechanisms, and the ability to operate at high temperature and high energy density [9]. These notable features indicate that heat pipe reactors present a promising option for use in terrestrial nuclear power generation, deep space exploration missions, and specialized military deployments [10].

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Nevertheless, it is notable that the honeycomb solid matrix utilized in heat pipe reactors differs from the conventional reactors that employ rod-shaped fuel and liquid coolant. Heat pipe reactors face several challenges, including limited heat transfer capacity under low-temperature conditions, difficulties in manufacturing and maintaining the complex monolithic core structure. Among these challenges, thermal stress induced by temperature fluctuations is particularly critical. Under normal operating conditions, temperature fluctuations in the monolith typically range from ±20 K to ±50 K, with maximum temperature differences reaching 108 K. During heat pipe failure scenarios, localized spikes can exceed 200 K. These fluctuations are exacerbated by the densely arranged fuel rods and heat pipe configuration, which increase the risk of exceeding thermal stress limits within the solid matrix, potentially leading to core damage or failure [11]. These issues, particularly thermal stress, represent significant challenges for the practical implementation of heat pipe reactor designs. Addressing them is crucial for ensuring the safety, reliability, and scalability of heat pipe reactors in future applications.

It has been demonstrated that heat pipe reactors have the potential to induce significant thermal stress when operated at peak capacity, especially in the event of a heat pipe failure. In their analysis report on the MegaPower project, Sterbentz et al. highlighted concerns regarding the potential for excessive thermal stress under operational and heat pipe failure conditions [12]. Ma et al. conducted further investigations into the generation of thermal stress under normal operating conditions and under conditions of heat pipe failure [13,14]. Li et al. conducted a detailed analysis of the variation in heat pipe wall temperature with respect to input power, demonstrating that at full power, the operating temperature of the heat pipe exceeds the material's yield strength [15]. Zhang et al.'s study on the MegaPower reactor further identified a critical issue: the peak stress could reach 169 MPa, surpassing the yield strength of the monolith material (131 MPa) at operational temperatures [16]. In the coupling analysis of the heat pipe segmented thermoelectric module converter (HP-STMC), Jiao et al. [17] demonstrated that unequal power distribution and suboptimal placement of heat pipes can result in elevated thermal stress within the heat pipe array in reactor designs where heat pipes and fuel elements are interleaved. The failure of the heat pipe will serve to exacerbate the problem of high stress levels. These findings underscore the imperative need for optimizing the thermal stress of heat pipe arrays.

Upon recognizing the issue of excessive stress, Sterbentz et al. proposed a series of solutions [18], including the utilization of stainless-steel tanks filled with liquid metal as the monolith, as well as the implementation of fuel-encased heat pipes. Moreover, adjusting the heat pipes to mitigate high stresses resulting from temperature differences is a feasible solution. Liu et al. explored the application of dual-ended heat pipes for effective temperature reduction, achieving significant thermal management improvements through optimized design and integration, thereby enhancing overall performance and reliability [19]. Jeong et al. achieved a significant reduction in stress by adjusting the temperature of the condensing section of the heat pipes [20]. However, these changes can affect the core's structural integrity and may not fully address heat pipe failure. Therefore, exploring parameter optimizations to reduce thermal stress while maintaining the current design is a viable approach.

As a solid-state reactor, heat pipe reactor is also distinguished by the strong and rapid interactions among various physical fields, including neutronics, thermal-hydraulics and mechanics. In particular, thermal expansion resulting from the temperature rise can significantly influence the distribution of fission power and the effective multiplication factor k_{eff} [21]. Moreover, the thermal expansion effect contributes to a significant portion of the negative feedback coefficient, which is significantly different than the feedback mechanisms in traditional nuclear reactors [22,23]. Consequently, the utilization of a multi-physics coupling approach is of paramount importance for the purpose of conducting a precise analysis and assessment of the system's performance.

When adjusting parameters to optimize reactor design, it is crucial not only to reduce the peak stress within the system but also to ensure that k_{eff} remains at the required level. This adjustment must take into account the inherent safety characteristics of the reactor core, such as negative feedback and temperature peaks, and must be executed without significantly compromising the overall safety of the reactor core. Therefore, the parameter optimization process necessitates multi-objective optimization under various constraints. Given these limitations and requirements, the optimization process must employ advanced algorithms for flexible and comprehensive multi-objective optimization, which inevitably involves substantial computational effort [24–26].

In multi-physics coupling computations, the computational time is extensively increased. The direct employment of numerical calculations for optimization on a large scale incurs a considerable expense, due to the substantial computational demands. Analysis of these calculations reveals multiple iterations with closely aligned parameters. Consequently, constructing a surrogate model, while ensuring accuracy, accelerates computation and reduces resource requirements [27], facilitating a more flexible and comprehensive optimization strategy. Recent advancements in machine learning techniques, such as deep neural networks and ensemble learning, have shown great potential in constructing high-fidelity surrogate models for complex multi-physics systems [28] [29]. These models not only reduce computational costs but also provide robust predictions under varying operating conditions, enabling more flexible and comprehensive optimization strategies.

Previously, our research team employed a multi-physics coupling numerical simulation method to study the distributions of temperature fields, stress fields, and neutron flux fields within heat pipe reactors [16]. Despite extensive simulations and validations, the coupling algorithm incurred high computational costs, and required improved accuracy and reliability under complex conditions. Additionally, practical engineering guidance was limited, necessitating further algorithm optimization.

In this paper, we propose a novel neural network-based surrogate model to replace traditional numerical simulations, significantly enhancing computational efficiency while maintaining accuracy. Unlike previous studies [30], which primarily focused on single-objective optimization or simplified thermal-hydraulic models, our approach integrates high-fidelity multi-physics coupling with ensemble learning techniques to achieve robust predictions under complex operating conditions. Comparative experiments verified the method's effectiveness, demonstrating good adaptability under complex conditions and providing better guidance for practical applications. Furthermore, we introduce a genetic algorithm-based multi-objective optimization framework that simultaneously balances thermal stress reduction, criticality maintenance, and safety margins, addressing a critical gap in current heat pipe reactor design.

The subsequent sections of the document are structured as follows: Section 2 presents the findings and identified issues based on numerical calculations conducted within this study, supported by the necessary theoretical groundwork. Section 3 discusses the practical implementation based on the established theories and models, along with the corresponding results. Finally, Section 4 provides a summary and discussion of these results.

2. Theory and methods

This section is organized as follows: Section 2.1 displays the framework of a multi-physics coupling analysis of a typical heat pipe reactor using numerical computation, identifying pertinent issues and suggesting directions for optimization. Section 2.2 details the transformation of the multi-physics coupling analysis framework from numerical computations into a data-driven model. Section 2.3 outlines the overarching structure of the data-driven model. Finally, Section 2.4 describes the multi-objective optimization strategy employed that accounts for the inherent conflicts in the optimization process.

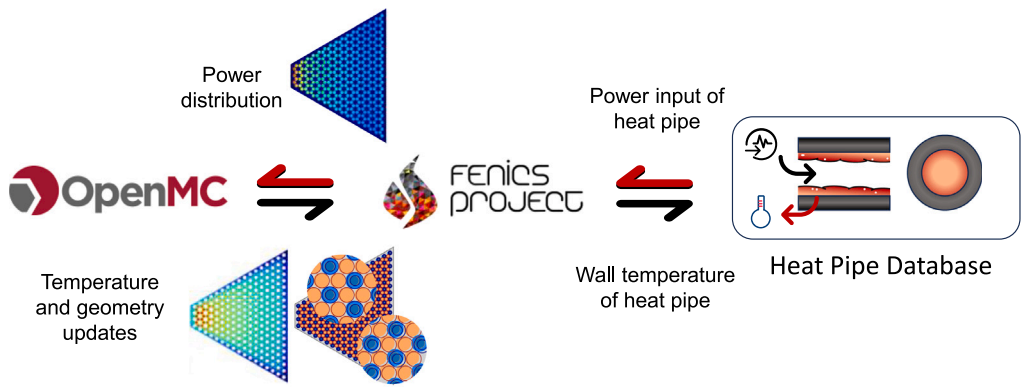

Fig. 1. A schematic of the multi-physics coupling framework.

2.1. High-order multi-physics coupling model

The schematic view of the multi-physics coupling framework is illustrated in Fig. 1. This framework integrates a neutron transport model using the Monte Carlo open-source computing platform OpenMC [31], a thermo-dynamic model using the finite element computing platform FEniCS [32], and a heat pipe wall temperature calculation model that ensures consistency between the heat pipe wall temperatures and the input power. To synchronize these distinct physical processes, we employ Picard iteration [33], a fixed-point method that iteratively solves the coupled nonlinear equations until convergence. It operates by sequentially decoupling the neutronics, thermal, and mechanical processes, starting with the initialization of the geometry and temperature fields. The neutronics model computes the power distribution and k_{eff}, which are then passed to the thermal model to solve the heat transfer equations and update the temperature field. The mechanical model subsequently calculates the thermal expansion and stress fields based on the updated temperature. Convergence is achieved when the maximum temperature difference between iterations falls below 1 K, with an under-relaxation factor of 0.1 applied to stabilize and accelerate the convergence process. This approach avoids the computational complexity of Jacobian matrix construction required by Newton–Raphson methods, making it more practical for large-scale multi-physics simulations. As the multi-physics coupling framework employs a high-fidelity description of the reactor core, we denote it as 'high-order' model in order for differentiation with the data-driven models described below.

As mentioned in the introduction, our previous research identified that the main factors contributing to high stress in heat pipe reactors are uneven input power at both ends of the heat pipes and excessive local temperature differentials caused by overly narrow heat transfer channels. Obviously, the high thermal stress can be addressed through optimizing the reactor design. To ensure compatibility, the preliminary optimization strategy aims to preserve the core characteristics of the original design as much as possible. Consequently, in devising solutions to mitigate stress, this study considers optimizing the original design by adjusting the core dimension, the fuel rod enrichment, and types of heat pipes. The heat pipe reactor MegaPower has been selected as the subject of study, as shown in Fig. 2.

Another issue exposed in the multi-physics coupling analysis framework is the prohibitive computational cost. Specifically, the computing time for a single computation reaches approximately six hours using 64 Intel CPU cores. Hence, the multi-physics coupling optimization using the multi-physics coupling analysis model becomes intractable. The computation times expended on different calculation processes are as shown in Table 1, with the most time-consuming ones being the neutronics calculation and the heat pipe wall temperature calculation, accounting for 66.7% and 27.4% of the total time, respectively. Therefore, the focus of this study is to reduce the computation times required by these two platforms, aligning them with the need for the reactor optimization.

Table 1
Time consumed by each process in the multi-physics coupling calculation of the heat pipe reactor.

Calculation process	Calculation time (min)	Time-consumption ratio (%)
Neutronics calculation	236	66.7
Monolith thermo-elastic calculation	9	2.5
Heat pipe wall temperature calculation	97	27.4
Duration of analysis	12	3.4
Total	**354**	**100.0**

2.2. Data-driven models using artificial neural networks

To speed up the multi-physics coupling analysis, the adoption of data-driven models to substitute for relevant numerical computations is considered. Data-driven neural networks inherently rely on the input of substantial datasets to train the network architecture [34]. The objective is to enable the network to recognize and learn the underlying patterns and regularities within the data. Artificial neural networks are composed of multiple artificial neurons [35], which represent the most fundamental unit of the network (see Fig. 3).

The synapses with weights $(\omega_{qi}, i = 1, 2, 3 \cdots)$ are responsible for receiving signals $(x_{qi}, i = 1, 2, 3 \cdots)$ transmitted from the neurons of the previous layer and weighting them before inputting them to the summation unit. The summation unit is responsible for linearly combining all weighted synaptic signals to produce output u_q, formulated as follows:

$$u_q = \sum_{j=1}^{n} \left(w_{qj} x_j - \theta_q \right) = w_q^T x \tag{1}$$

A regression neural network [36] is a data-driven approach capable of predicting continuous output values based on input data, with strong nonlinear fitting capabilities. The activation function $f(\cdot)$ is generally a linear function responsible for applying transformation to the output. The threshold/bias θ_q adjusts the input to the activation function by subtracting/adding this value. Upon transformation by the activation function, the final output of the neuron, denoted as y_q, is given by:

$$y_q = f(u_q) = f\left(\sum_{j=1}^{n} w_{qj} x_j - \theta_q \right) \tag{2}$$

In this context, LeakyReLU is selected as the activation function. This choice is made to circumvent the issue of neuron death that may occur during training with the standard ReLU. The mathematical expression for LeakyReLU is as follows:

$$\text{LeakyReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ \alpha x & \text{if } x \le 0 \end{cases} \tag{3}$$

To fulfill the requirement for rapid multi-physics field prediction, after thorough consideration of the characteristics of various neural

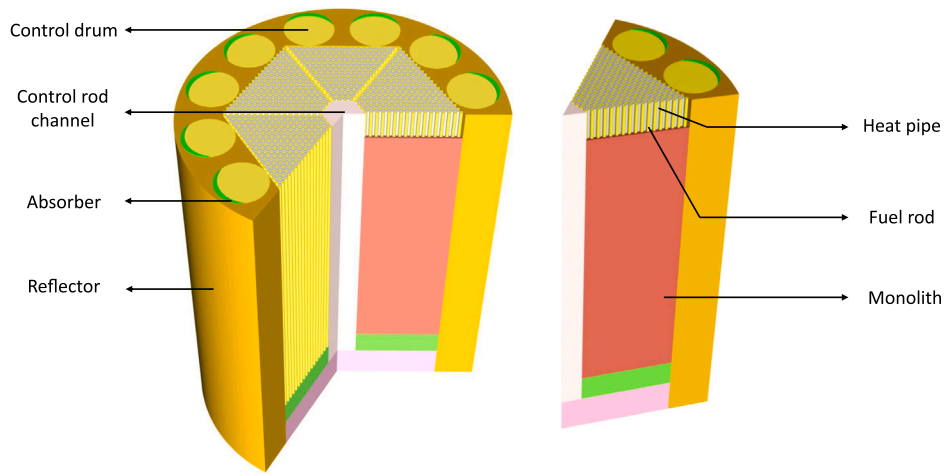

Fig. 2. Schematic of the MegaPower neutronic model. The reactor components include the stainless steel monolith (red), 19.75% enriched UO_2 fuel rods (gray), and sodium heat pipes (yellow). Reactivity is controlled by a cylindrical disk (gold) located within the alumina reflector layer (orange), which rotates to position the crescent-shaped B_4C absorber (light green) towards the core.

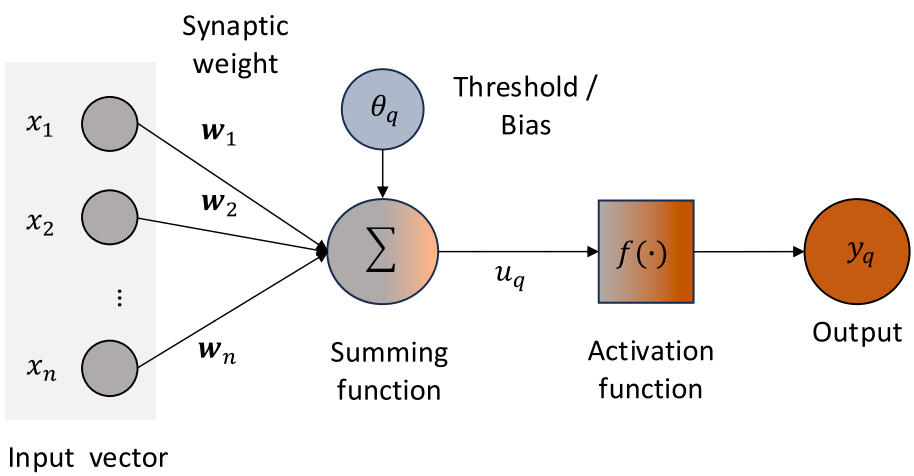

Fig. 3. Schematic of the artificial neuron model.

network models, a fully connected deep neural network, as illustrated in Fig. 4, has been selected as the data-driven model. The depicted network architecture includes an input layer, three hidden layers, and an output layer. The numbers of neurons in the input and output layers are determined by the dimensions of the input and output data, respectively. The configuration of the hidden layers — including the number of layers and the number of neurons per layer — is dynamically adjusted based on the complexity of the specific problem at hand, thereby optimizing the network's fitting performance. To fulfill the requirement for rapid multi-physics field prediction, we conducted a thorough evaluation of various neural network architectures and selected a fully connected deep neural network as illustrated in Fig. 4 as the data-driven model. The network consists of an input layer, three hidden layers, and an output layer, with the number of neurons in each layer determined by the dimensions of the input/output data and the complexity of the problem. The choice of three hidden layers was based on a systematic hyperparameter optimization process using Keras Tuner, which balances model complexity and predictive accuracy [37]. This architecture effectively captures the nonlinear relationships in the data while maintaining computational efficiency, making it well-suited for large-scale optimization tasks [38].

During the model training phase, the loss function is selected to quantify the discrepancy between the neural network's current output and the desired target output. The model parameters, including weights and biases of neurons, are iteratively updated through the back-propagation algorithm [39] employing stochastic gradient optimization. This study utilizes Mean Squared Error (MSE) loss functions prevalent in regression neural networks:

$$L_{\mathrm{MSE}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \|y_i - \hat{y}_i(\theta)\|_2^2 \tag{4}$$

In this formulation, $\|\cdot\|_2$ denotes the L_2 norm, where y_i represents the value vector for the ith sample, and $\hat{y}_i(\theta)$ is the predicted value vector for the ith sample using the model parameters θ.

Furthermore, tailored to the requirements of different problems, additional loss functions such as the Mean Absolute Error (MAE) and the Mean Absolute Percentage Error (MAPE) are also employed [40].

$$L_{\mathrm{MAE}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \|y_i - \hat{y}_i(\theta)\|_1 \tag{5}$$

$$L_{\mathrm{MAPE}}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \left\| \frac{y_i - \hat{y}_i(\theta)}{y_i} \right\|_1 \cdot 100\% \tag{6}$$

Here, $\|\cdot\|_1$ signifies the L_1 norm, which is the sum of the absolute values of the elements in the vector [41].

Additionally, the Huber loss [42] is also considered:

$$L_{\delta}(\theta) = \frac{1}{N}\sum_{i=1}^{N} \phi_{\delta}(y_i - \hat{y}_i(\theta)) \tag{7}$$

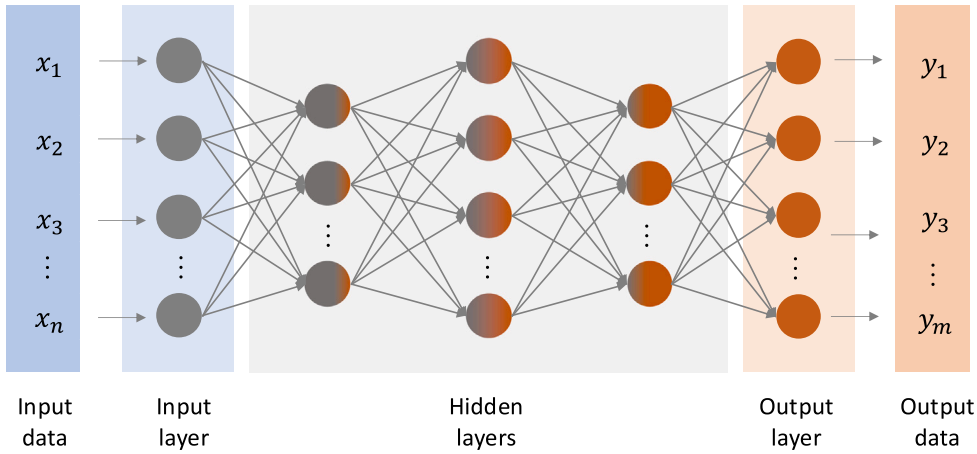

Fig. 4. Schematic of deep neural network.

$$\phi_\delta(a) = \begin{cases} \frac{1}{2}a^2 & \text{if } |a| \le \delta \\ \delta|a| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \tag{8}$$

During the training process, the Adam algorithm [43] is employed as the optimization technique. This algorithm incorporates the concepts from both Momentum and RMSprop (Root Mean Square Propagation) optimization methods [44,45]. The Adam algorithm adaptively adjusts the learning rate for each parameter by computing estimates of the first-order moments (i.e., the mean of the gradients) and the second-order moments (i.e., the uncentered variance of the gradients). The key aspects are as follows:

Initially, the parameter vector θ_0, the first-order moment vector m_0, the second-order moment vector v_0, and the time step t are initialized. Iteration proceeds until the parameter vector θ_t converges:

$$t = t + 1 \tag{9}$$

$$g_t = \nabla_\theta f_t(\theta_{t-1}) \tag{10}$$

Here, g_t is the gradient of the loss function with respect to the parameter vector, and the training target is to minimize the expected value of this function with respect to the parameters θ.

$$m_t = \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t \tag{11}$$

$$v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2 \tag{12}$$

The above steps yield the first-order and second-order moments, where β_1 and β_2 are the exponential decay rates for the first-order and second-order moments, respectively. These rates control the exponential decay of the moving averages, which decrease with each step. Subsequently, a bias correction is applied to both the first-order and second-order moments, yielding $\hat{m}_t$ and $\hat{v}_t$:

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \tag{13}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \tag{14}$$

$$\theta_t = \theta_{t-1} - \alpha \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \tag{15}$$

This results in the updated parameter vector θ_t, where α represents the learning rate, and ϵ is a small value, set to 0.01, introduced to prevent division by zero.

2.3. Multi-physics coupling data-driven model

The data-driven model is established by substituting each numerical computation process using its data-driven model. Specifically, six data-driven models are trained and coupled, using datasets generated by OpenMC and FEniCS. Fig. 5 illustrates the workflow of the data-driven model.

OpenMC and FEniCS are conceptualized as black boxes that generate data based on input information, and the output data are categorized to form five neural network models and one database. Five of these networks are further merged into two larger models to substitute the numerical computation platforms. Specifically, OpenMC is utilized to produce the power distribution field and the k_{eff}, while FEniCS generates the monolith temperature field, displacement field, and temperature gradient field.

The input and output specifications for each model are listed in Fig. 6. The coupling relation between the models in the data-driven model is as follows:

Step 1: Generate initial input parameters for the neutronics model, the monolith thermo-elastic model, and the heat pipe model.

Step 2: The neutronics model is used to calculate k_{eff} and power distribution, subsequently updating the power distribution.

Step 3: The heat pipe model is used to generate the input power for each heat pipe based on the input parameters. Subsequently, the heat pipe database generates the wall temperature distribution for each heat pipe according to the input power. If the wall temperature distribution converges, the iteration proceeds to the next step; if not, the heat pipe wall temperature distribution is updated, and step 3 is repeated.

Step 4: The monolith thermo-elastic model uses the input parameters to calculate the average temperature of the monolith, the edge temperature of the fuel, and the thermal expansion conditions. If all fields have converged, combined with calculations of fuel temperature, this model is capable of delivering information on the average and peak temperatures of each fuel rod, thereby yielding data on the highest and average temperatures across all fuel rods; otherwise, the input parameters for displacement, temperature, and density are updated, and the process returns to step 2.

Step 5: The resulting output is generated and the data-driven model of the multi-physics fields is terminated.

2.4. Multi-objective optimization method

During the optimization process, multiple competing objectives should be considered simultaneously. Generally, these objectives are integrated into a composite utility function to form a unified optimization goal. However, the utility function adopted may not optimally combine these objectives a priori. Recognizing this, the present study adopts

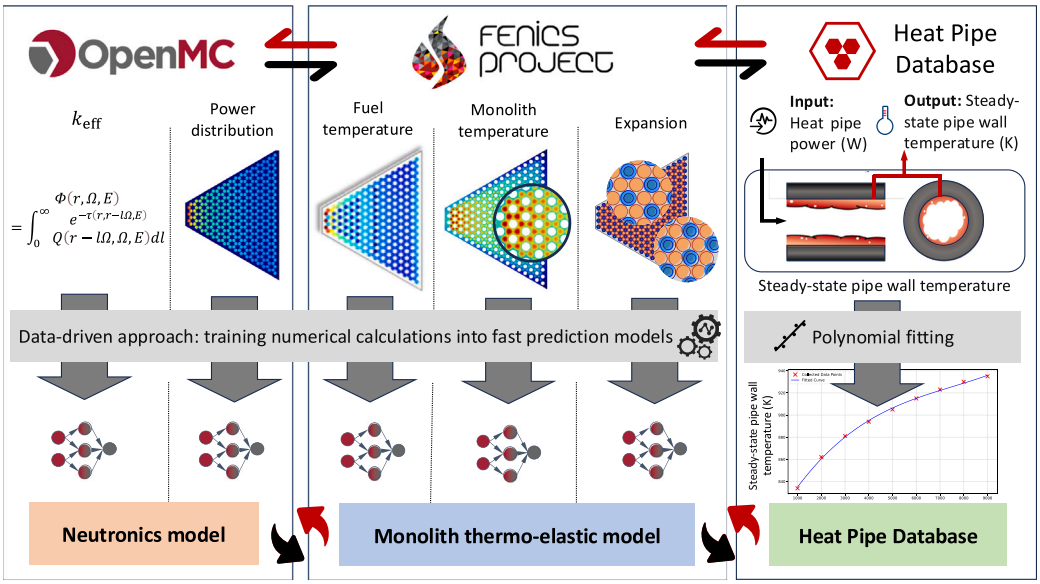

Fig. 5. The workflow of the multi-physics data-driven model.

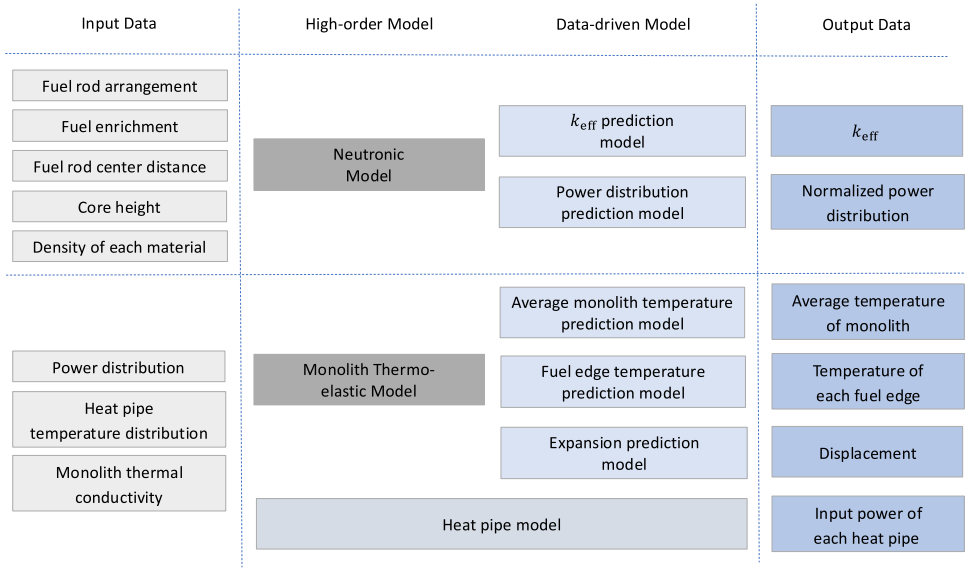

Fig. 6. Basic information for each model.

the multi-objective optimization methods to effectively reconcile the competition among these objectives. We employ Non-dominated Sorting to obtain the Pareto Frontier [46], which facilitates multi-objective optimization. The dominance relationships and the Pareto frontier are illustrated in Fig. 7.

Within the same non-dominated frontier, transitioning to different solutions usually involves a trade-off, sacrificing performance in certain objectives to gain benefits in others. Therefore, the Pareto optimal set often comprises multiple non-inferior solutions. This enables the selection of the most suitable solution and significantly enhancing the flexibility of decision-making, by quantifying the trade-offs based on specific circumstances [47].

In this work, non-dominated solutions are generated using the Non-dominated Sorting Genetic Algorithm-II (NSGA-II) [48,49]. Fig. 8 presents the specific flowchart of NSGA-II. Specifically, modifications to the standard NSGA-II are performed, by limiting duplicate genes to further promote population diversity, and adopting dynamic programming to accelerate the fitness generation process. In addition to using non-dominated solutions to replace the selection process of genetic algorithms, optimizations specific to the genetic algorithms are also incorporated. These include the introduction of an elitism strategy to ensure the retention of optimal individuals and the introduction of crowding distance to maintain population diversity [50–52].

The elitism strategy [53] is implemented by placing offspring and parents in the same selection pool in each selection phase. An individual's crowding distance is defined by calculating the distance between the individual and its neighboring entities within the normalized objective function space. The computation scheme is depicted in Fig. 7, where the crowding distance of the ith solution can be represented by the side length of the dashed-line rectangle. In cases where there are no neighboring solutions on one side, the crowding distance is set to infinity. Among two solutions on the same non-dominated front,

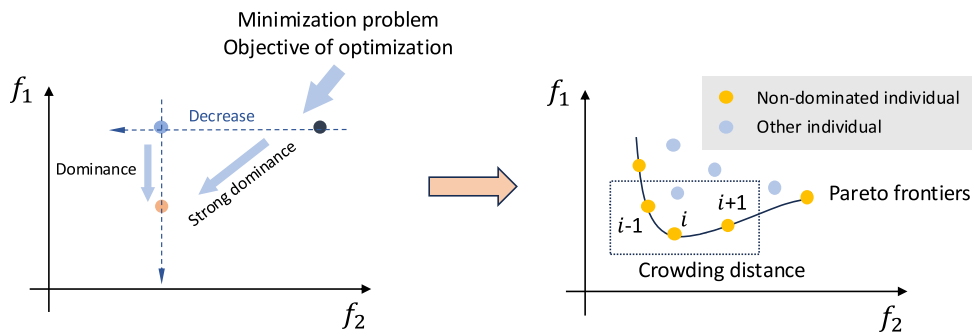

Fig. 7. Dominance relations (left) and Pareto frontiers with crowding distance (right).

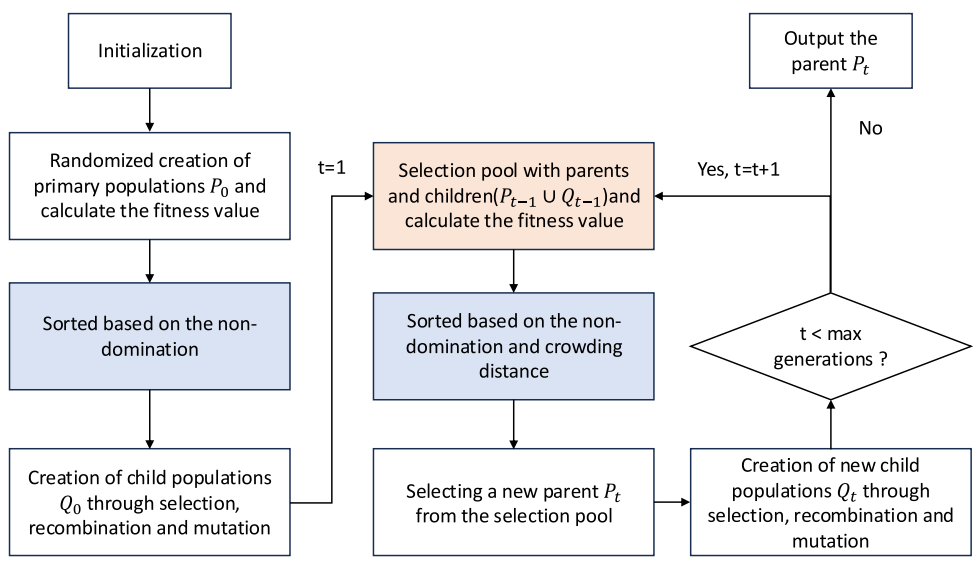

Fig. 8. NSGA-II algorithm flowchart for multi-objective optimization. Blue section: generating non-dominated solutions; Orange section: elite strategies.

the one with the greater crowding distance is preferred, as it represents a sparser region within the population, thus contributing to the population diversity.

3. Results and analysis

In this section, we delineate the training of the data-driven model and the optimization of the heat pipe reactor utilizing the data-driven model and the multi-objective optimization method as described in Section 2. The generation and segmentation of the dataset is described in Section 3.1. Section 3.2 details the training of the neural-network-based data-driven model utilizing the dataset. In Section 3.3, the multi-physics field data-driven model is applied to the heat pipe reactor, and its accuracy is verified by comparing it with high-order numerical computation results. Section 3.4 illustrates the initialization of the genetic algorithm and the heat pipe selection, conducted after the determination of the specific data-driven model and the optimization requirements. Lastly, Section 3.5 demonstrates the optimization process and its results.

3.1. Generation of training data

During the network training phase, the accuracy of the model relies on a dataset that is both diverse and of high quality. The dataset for neutron transport calculations was generated using OpenMC. Considering the need for optimization and the requirements of multi-physics coupling, the input features include the location label for low and high enrichment fuel rods, the specific enrichments of both high-enriched and low-enriched fuels, the active core height, the core displacement, the average temperature of fuel rods, the average temperature of heat pipes, the average temperature of monolith, and the densities of the fuel and monolith. The selection of these parameters aims to ensure that the model covers the possible variations in core parameters, considering the feedback effects of temperature and displacement. The primary output variables include the core k_{eff} and the power distribution of the fuel rods. In order for not changing the core size, the total number of fuel rods is fixed according to the original design, resulting in a number of ten rows of fuel rods in the core. Each row of the fuel rods can select between two labels “high-enrichment” or “low-enrichment”. The enrichment for the “high-enrichment” label is set as ranging from 16% to 19.75%, and the enrichment for the “low-enrichment” label can vary ranging from 8% to 16%. The active core height can be selected ranging from 140 cm to 160 cm. The fuel rod pitch can be selected ranging from 1.6 cm to 1.7 cm. The density values for both the fuel and the monolith range from 85% to 100% of their original densities (see Fig. 9).

In the interest of reducing the training dataset as well as simplifying the training process, the temperature distribution and the displacement distribution are updated according to the core-averaged values. This simplification resulted in negligible power errors of less than 0.15% and k_{eff} error of less than 100 pcm. This is mainly attributed to the long neutron mean-free-path of the reactor caused by the fast neutron spectrum. Based on this error analysis, it is considered that using average values for temperature and displacement updates falls within an acceptable range of impact on neutron transport calculations.

The dataset for the thermal–mechanical coupling calculations was generated using FEniCS, with input variables being the heat pipe wall

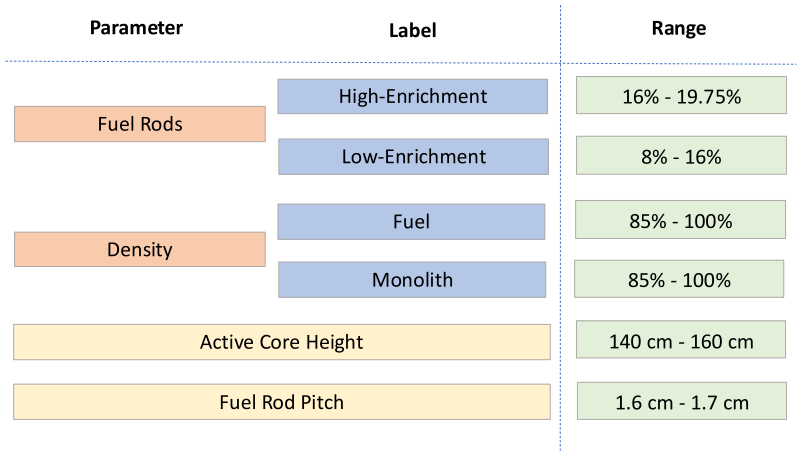

Fig. 9. Parameters being optimized and their variation ranges.

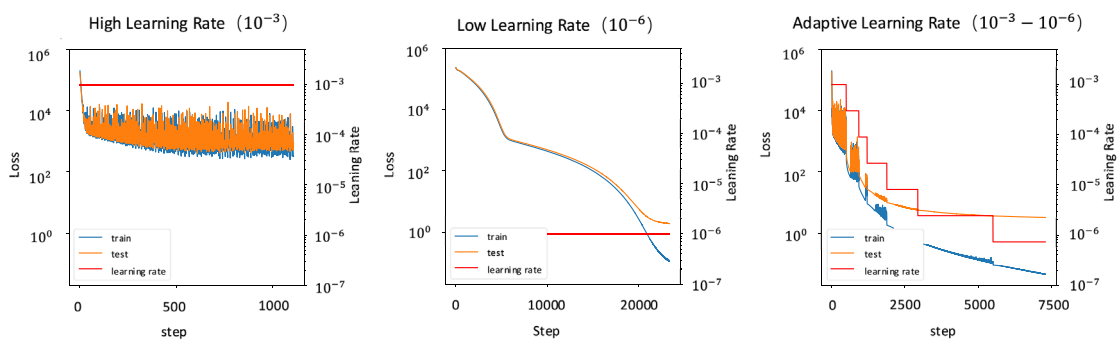

Fig. 10. The effect of different learning rates on training. The red line represents the learning rate, the blue line represents the loss function for the training set, while the orange line indicates the performance on the validation set. The scenarios depicted include fixed learning rates with both high (left) and low (middle) values, as well as the training condition with an adaptive learning rate (right).

temperatures and the averaged fuel rod power. The output variables comprise the input power to the heat pipes, the edge temperature of the fuel, the average temperature of the monolith, and the average expansion rate of the monolith. A total of 1700 neutron transport simulation data and 20,000 thermal–mechanical coupling simulation data are generated. During the model training phase, the datasets were divided into training, validation, and test sets in a ratio of 7:2:1.

3.2. Training of the data-driven model

A multitude of strategies has been adopted to improve training efficiency as well as to enhance the data-driven model's fitting capabilities and generalizability. Taking the training of the power distribution prediction model as an example, the adaptive learning rate [54] technique and the early stopping technique [55] are employed. These two techniques significantly accelerated the training process and endowed the model with greater flexibility. The adaptive learning rate strategy incorporates a decay coefficient to monitor the loss function on the training set. If no improvement is detected within a designated number of epochs, the learning rate is reduced by multiplying it with the decay coefficient. In the early stage of the training process, the adaptive learning rate strategy results in rapid convergence through a high learning rate, preventing the model from falling into gradient plateaus. In the late stage of the training process, the adaptive learning rate strategy ensures model stability and enhanced generalization performance by using a low learning rate. Fig. 10 compares the training performance with and without the adaptive learning rate on the k_{eff} prediction model.

In order to improve the prediction accuracy with limited datasets, this study has utilized the ensemble learning strategy tailored for predicting the power distribution [56]. Maintaining the ratio between the training and test sets, different neural network architectures were employed to generate three different data-driven models. It is evident that the application of ensemble learning has significantly reduced the maximum errors. 43% of cases exceeded an error of 1% when single neural network model was applied, while this proportion dropped to 4% with the ensemble model.

The neutronic data-driven model replacing the OpenMC calculation is comprised of a k_{eff} prediction model and a power distribution prediction model. During the training of the k_{eff} prediction model, MAPE loss function was utilized. The model exhibits prediction errors contained mostly within 200 pcm, and an average prediction error of around 50 pcm. As an analytical tool for assessing the criticality of the heat pipe reactor, such an accuracy level for k_{eff} is within a satisfying range.

The monolith thermal-elasticity model acts as an alternative to the FEniCS calculation. This data-driven model is composed of three sub-models: the average monolith temperature prediction model, the fuel edge temperature prediction model, and the expansion prediction model. The average monolith temperature prediction model contributes to the temperature feedback mechanism. Mean Squared Error (MSE) was chosen as the loss function during the training process. The average monolith temperature prediction errors are well confined within 1 K. Such a level of accuracy ensures that the model can provide precise temperature feedback for neutronics simulations, meeting the requirements of multi-physics coupling analysis.

The fuel edge temperature prediction model is devised for generating the average temperature values at the fuel rod edges. The average and peak temperatures of each fuel rod serve as a critical dataset for the temperature feedback mechanism. The MAE loss function which

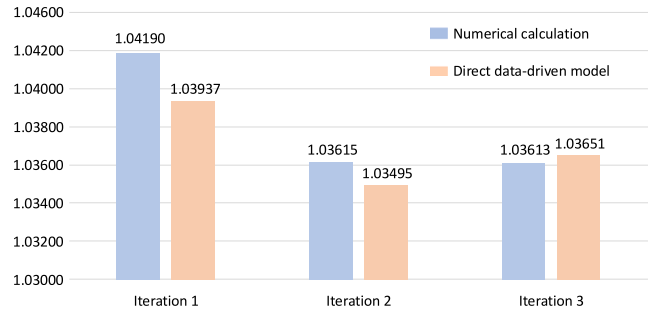

Fig. 11. Variation of k_{eff} with iteration steps.

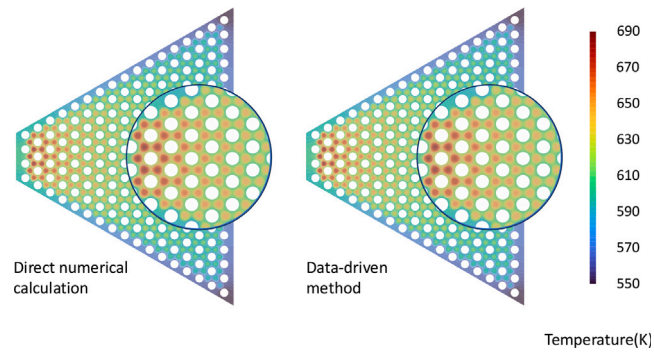

Fig. 12. Comparison of temperature distribution.

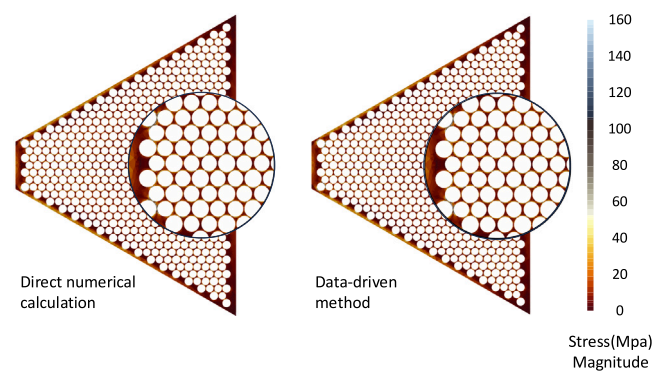

Fig. 13. Comparison of stress distribution.

measures the difference between predicted values and actual target values, is utilized to assess predictive performance. The fuel temperature prediction model yields an average error of less than 0.2 K. For the temperature prediction of individual fuel rods, the maximum error is controlled within 10 K.

The expansion prediction model focuses on evaluating the geometrical changes in the monolith boundaries due to the thermal expansion of the core. These changes, being relatively small compared to the overall boundary dimensions of the monolith material, might be overlooked during the training process. Therefore, a scaling strategy is adopted to improve the model's sensitivity to the subtle expansion changes at the monolith boundaries. The expansion changes are extracted as separate quantities from the overall boundary data, isolating the minor expansions from the larger dimensions of the monolith. To ensure these small changes are accurately represented in the model's predictions, a scaling factor is applied to the extracted change quantities. This factor amplifies the minute variations, making them more prominent during the training process and improving the model's fitting accuracy. MSE was chosen as the loss function during the training of this model. The expansion prediction model demonstrates an average error of less than 2%, a level of accuracy that provides accurate displacement feedback and meets the accuracy requirements for data-driven model.

3.3. Multi-physics coupling analysis using the data-driven model

Utilizing the data-driven model as described in the previous section, a steady-state multi-physics field coupling analysis was conducted on MegaPower, and the results were compared against those obtained from high-order multi-physics coupling frameworks. Compared to the high-order calculations, the computational time was significantly reduced from 6 h to 4 min. Fig. 11 illustrates the variation of the k_{eff} with the number of iteration steps for both the data-driven model and numerical calculations. The k_{eff} obtained from data-driven model showed a systematic error of around 200 pcm from the numerical calculations. It should be emphasized that this discrepancy remained essentially constant with iteration, indicating that the data-driven model accurately reflects the influence of other physical fields on k_{eff}. Furthermore, the average fuel temperature and fuel pitch obtained from data-driven model were in close agreement with the numerical results, with the average fuel temperature discrepancy being within 3 K and the monolith displacement discrepancy being below 0.1%.

Figs. 12 and 13 illustrate the monolith temperature distribution and the monolith stress distribution obtained by the data-driven model those calculated using high-order models. Note that the temperature and stress distributions for the data-driven model were generated directly. They are calculated by thermal-elasticity numerical solvers using thermodynamic boundary conditions provided by the data-driven model. It can be found that the results from data-driven model closely matched those from high-order calculations.

The final values obtained from data-driven model were: k_{eff} = 1.03113, fuel pitch expansion rate of 1.00981, average fuel temperature of 920 K, average monolith temperature of 897 K, peak temperature of 952 K, and peak stress of 169.2 MPa, with a total computation time of 216.6 s. In contrast, direct numerical calculations yielded: k_{eff} = 1.03621, central expansion rate of 1.00983, average fuel temperature of 923 K, average monolith temperature of 897 K, peak temperature of 952 K, and peak stress of 168.47 MPa, with a total computation time of approximately 360 min (see Table 2).

Table 2
Comparison between data-driven models and direct numerical calculations under steady-state multi physics coupling calculations.

Parameter	Data-driven model	Direct numerical calculation
k_{eff}	1.03113	1.03621
Fuel pitch expansion rate	1.00981	1.00983
Average fuel temperature (K)	920	923
Average monolith temperature (K)	897	897
Peak temperature (K)	952	952
Peak stress (MPa)	169.2	168.47
Total computation time (s)	216.6	21 600

By benchmarking against high-order multi-physics coupling analysis, the proposed method significantly increases the efficiency of analysis by reducing the time for a single analysis from hours to minutes, while maintaining sufficient calculation accuracy. Consequently, in scenarios where detailed field distributions are not necessary, the data-driven model can be utilized to accomplish the multi-physics coupling analysis tasks. This approach enables the capability to carry out extensive computations required during the reactor optimization.

3.4. Multi-objective optimization of the nuclear reactor

The optimization via genetic algorithms aims to identify the optimal core layout, core geometry and fuel enrichment to minimize the peak stress while maintaining the criticality and safety features

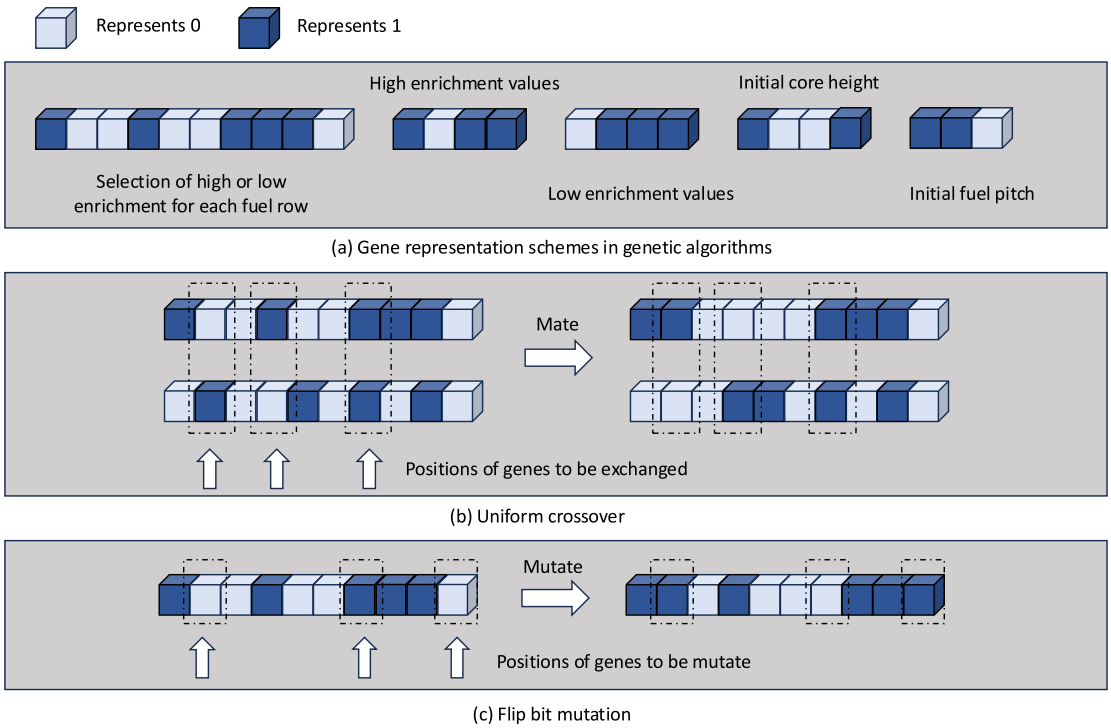

Fig. 14. (a) Gene representation schemes in genetic algorithms, illustrating the encoding of solutions; (b) Uniform crossover, demonstrating the random exchange of genes between two parent solutions; (c) Flip bit mutation, showing the process of altering a gene's value to explore new solutions.

of the reactor. To this end, the optimization problem is abstracted into a genome composed of a 25-bit binary string. Fig. 14 provides a schematic representation of the genomic structure, in which the first ten digits corresponds to the enrichment labels of the ten rows of fuel rods within the reactor core, ordered from the inner to the outer core. Each bit of this ten-digit sequence indicates whether the corresponding row of fuel rods employs low-enrichment fuel.

The subsequent eight binary digits are divided evenly into two sets, each representing the enrichment levels for high-enrichment and low-enrichment fuels, respectively. This quartet segmentation ensures that each enrichment level can be chosen among 16 different grades. The last seven binary digits are allocated such that the first four can be used to represent the core height, and the remaining three can be used to denote the fuel rod pitch. This encoding strategy is designed such that the design parameters can be fine-tuned to search for the optimal settings. During the construction of the data-driven model, the training datasets are generated in ranges which cover entirely the space that the optimization parameters can possibly be chosen in.

In the genetic algorithm framework, three fitness criteria are considered for optimization: minimizing the global peak stress (termed Fitness 1), minimizing the average of the top 30% of maximum stress values in each region (referred to as high-stress values, termed Fitness 2), and maximizing the k_{eff} (termed Fitness 3). Fitness 2 is incorporated to reduce the overall thermal stress in the reactor core. In addition to the optimization target of the peak stress and the k_{eff}, the high-stress values are also incorporated into the genetic algorithm's fitness function. This inclusion aims to prevent scenarios in which optimizing for peak stress in one area inadvertently leads to increased stress in other regions, thereby elevating the risk to the reactor core.

Directly calculating the stress for every individual in the population would be computationally intensive and time-consuming. By using temperature differences, which are easier and faster to compute, the algorithm can rapidly evaluate and compare different solutions. To improve the optimization efficiency, the peak temperature difference between adjacent heat pipes is utilized as a proxy for the peak stress in that area. Following optimization, specific stress data will be determined through numerical calculations. Besides, to ensure the safety of the reactor design, a negative temperature coefficient must be guaranteed. If an individual does not meet these safety standards, its fitness values across all three criteria are artificially degraded to facilitate its elimination from the population.

During the genetic crossover process, a uniform crossover operator is employed. All genes within the genome are exchanged with a probability of p allowing for effective exploration of the genetic space while maintaining efficient information exchange. This probability p is dynamically adjusted based on the characteristics of the problem, ensuring an optimal balance between exploration and exploitation. Considering the binary representation of the existing genomes, a bit-flip mutation operator is chosen for gene mutation. Each gene produced by the crossover operator is assessed for a potential bit-flip based on a given probability, thereby further expanding the search space.

As is mentioned above, the critical issue of the current reactor concept lies in the excessive thermal stress in the monolith, even in the normal operation state. Since the thermal stress is proportional to the temperature gradient, we propose to deploy different types of heat pipes in different locations, instead of using a single type of heat pipe throughout the reactor core. As will be shown, with the aid of the multi-objective optimization method, this strategy is particularly beneficial for reducing the temperature gradient and the peak thermal stress.

In choosing the heat pipe types, we propose to tune the inner wall thickness while maintaining the overall heat pipe dimension, as this is deemed a strategy that facilitates the manufacturing and installation of heat pipes. However, other strategies effective in changing the heat lifting capacity can be adopted, as long as they are proven to be practical. Starting from an original wall thickness of 2 mm, the heat transfer efficiency of the heat pipes is gradually reduced by increasing the wall thickness, resulting in different wall temperatures with the same input power. These varying thicknesses of heat pipes are categorized in ascending order from $type0$ to $type5$, with $type0$ corresponding to a wall thickness of 2 mm, $type1$ to 2.5 mm, $type2$ to 3 mm, $type3$ to 3.5 mm, $type4$ to 4 mm, and $type5$ to 5 mm.

During the study, the heat pipe wall temperatures were calculated using the heat pipe analysis code. The calculation employs input powers

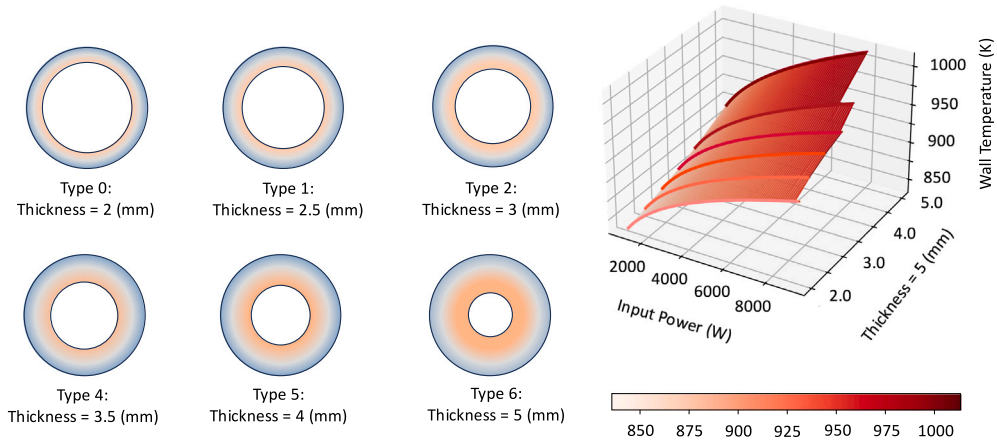

Fig. 15. Input power-wall temperature curves for heat pipes with different wall thicknesses.

ranging from 1000 W to 8000 W using an interval of 100 W. A total of 420 data points were collected, leading to the creation of a database that encapsulates the relationship between input power and wall temperature for the 5 heat pipe types. Fig. 15 presents the wall temperature versus input power for heat pipes with different wall thicknesses. This database provides an effective foundation for the adjustment of heat pipe types in subsequent optimization processes.

In practice, the optimization of heat pipe types is embedded within the genetic algorithm optimization framework, by iteratively refining the heat pipe configuration based on temperature differential feedback. the genetic algorithm begins by generating an initial population of solutions based on various parameters. During the evaluation phase for each individual solution, the temperature differential between adjacent heat pipes is calculated, guiding adjustments to the heat pipe types to minimize these differentials. This step is crucial for achieving a uniform temperature distribution, thereby reducing thermal stress. After adjusting the heat pipe types, fitness scores for each individual are generated, incorporating the effects of these adjustments on the overall thermal profile. The genetic algorithm then uses these fitness scores to guide the evolution of the population.

3.5. Optimized results

This section describes the optimization effects of the reactor based on the proposed data-driven model and genetic algorithms. To identify the specific effects of heat pipe type optimization, we compare the optimized results using a single type of heat pipe and those by utilizing different heat pipe types. Comparisons of the different processes and results are depicted in Figs. 16 and 17. The optimization using one type of heat pipe yields optimal results after a few iterations, but the improvements in the stress distribution are not significant. Moreover, the optimization process for fitness 3 (i.e., k_{eff}) shows an average value remained below 1, indicating that the core cannot achieve criticality using this optimization strategy. This defect can be attributed to the attempt that the optimizer is trying to alleviate the excessive stress by reducing the fuel enrichment. Eventually, an individual with a k_{eff} greater than one and the lowest stress was selected as the final optimized reactor concept. In this concept, the peak temperature difference distribution map for adjacent heat pipes showed that while there was some improvement by adjusting the core parameters, the large temperature difference between the heat pipes and their neighbors persisted in the peripheral region of the core.

As is shown in Fig. 16, both stress and k_{eff} showed satisfactory optimization performance. The stress values were reduced to a much lower level, and the core criticality is ensured. Hence, it can be concluded that the optimization of heat pipe types added significant flexibility to the optimization process, allowing for the thermal stress to be minimized with minimal impact on k_{eff}. An individual with both excellent stress performance and k_{eff} was ultimately selected as the optimized reactor concept.

The optimized results show that the optimization without considering the heat pipe type yields a reduction of the thermal stress value from 169 MPa to 129 MPa. Although this stress value is slightly lower than the material's yield tensile strength at high operating temperatures (131 MPa), there remains a potential safety risk, let alone that the reduction comes at a considerable sacrifice on k_{eff}. After incorporating adjustments to heat pipe types, the peak stress value was further reduced to 68 MPa, which is 40% of the stress of the original concept. Moreover, this is achieved by a minor decrease in k_{eff} relative to the original design. At this point, the peak stress value is approximately 51% of the material's yield tensile strength at high operating temperatures (131 MPa). These results indicate that the peak stress within the reactor has been effectively controlled, ensuring the core integrity and increasing the safety margin of the reactor.

4. Conclusion and discussions

This study introduces an advanced framework for optimizing the MegaPower heat pipe reactor by integrating a data-driven surrogate model with multi-objective optimization techniques. Utilizing a neural network-based surrogate model, the simulation time was significantly reduced from approximately six hours to four minutes while maintaining high prediction accuracy. The discrepancy in k_{eff} between the surrogate model and the direct numerical calculation was within 200 pcm, with peak temperature and stress differences of 3 K and 1.9 MPa, respectively.

Key findings from the high-fidelity multi-physics coupling simulations include the significant impact of thermal expansion on core reactivity and structural stress. The study revealed that at full power operation, thermal expansion decreased material density and increased fuel spacing, resulting in a reduction of the negative temperature feedback coefficient by 537 pcm. Additionally, due to the non-uniform heat pipe wall temperature, the temperature difference within the reactor core increased, with fuel rod temperature differences rising from 34 K to 93 K and matrix temperature differences increasing from 34 K to 108 K. In the event of heat pipe failure, the analysis indicated that peak stress within the core could exceed the yield strength of stainless steel at operating temperatures, posing a risk to the reactor's structural integrity. In scenarios involving three heat pipe failures, the peak stress was predicted to exceed the ultimate tensile strength of stainless steel, potentially leading to core damage.

To optimize the core structure design, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) was employed for multi-objective optimization, effectively balancing multiple objectives such as minimizing

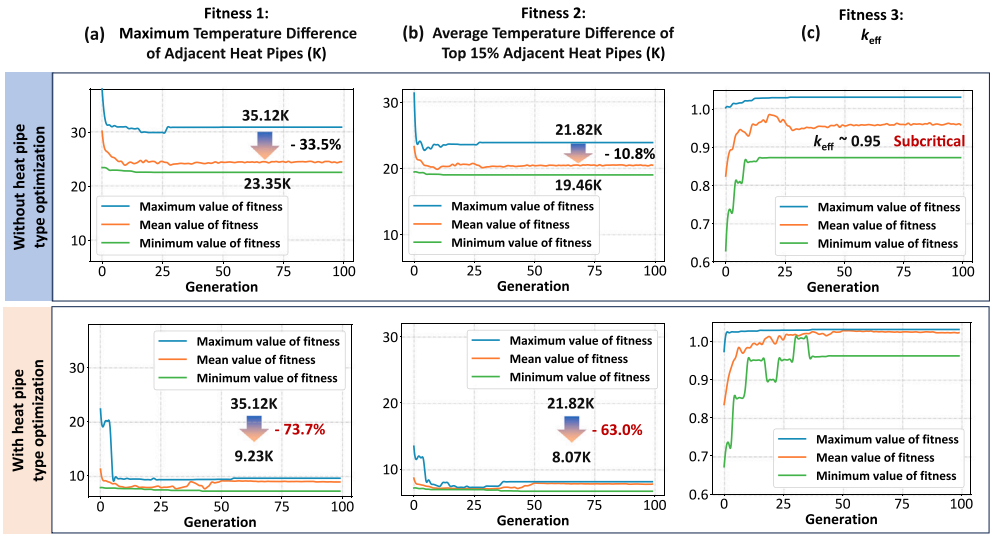

Fig. 16. The optimization processes are delineated as follows: Process (a) signifies the optimization of the highest stress (i.e., fitness 1), Process (b) corresponds to the optimization under average stress difference of top 15% adjacent heat pipes, denoted as high stress (i.e., fitness 2), while Process (c) pertains to the optimization of the k_{eff} (i.e., fitness 3). During these optimization processes, the red curve represents the highest fitness value within the population, the orange curve denotes the average value, and the green curve indicates the lowest value.

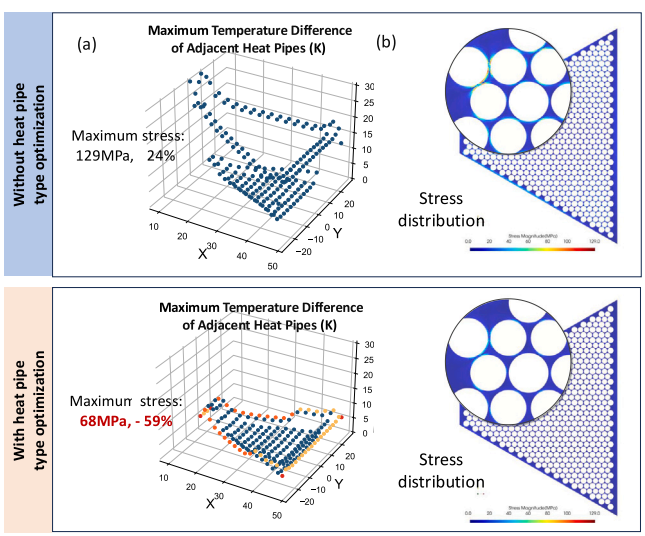

Fig. 17. The optimization outcomes are delineated as follows: (a) represents the peak temperature and corresponding stress differential between adjacent heat pipes, (b) represents a partial stress distribution image resultant from the optimization process.

thermal stress and ensuring critical safety parameters. Through this optimization, the peak stress within the core was reduced by 59%, from 169 MPa to 68 MPa, thereby enhancing the structural integrity and safety margin of the reactor.

The introduction of the surrogate model and optimization framework not only improved computational efficiency but also provided a reliable tool for real-time core optimization and safety analysis. This framework demonstrated robust adaptability to complex operating conditions, ensuring reliable predictions for both normal operation and failure scenarios.

Overall, the developed framework represents a significant advancement in the predictive modeling and optimization of heat pipe reactors, paving the way for more efficient and reliable reactor designs. Future work will focus on further refining the surrogate model, achieving a more independent selection of high and low enrichment levels, and enhancing the burnup function and control drum rotation scenario calculations based on the identified core parameters.

CRediT authorship contribution statement

Junda Zhang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Yinuo Chen:** Writing – review & editing, Writing – original draft, Visualization. **Xiaojing Liu:** Writing – review & editing, Project administration, Funding acquisition. **Xiang Chai:** Writing – review & editing. **Hui He:** Writing – review & editing. **Jinbiao Xiong:** Writing – review & editing. **Tengfei Zhang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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